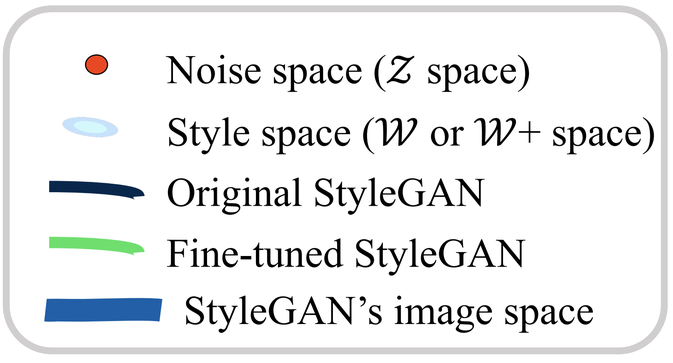
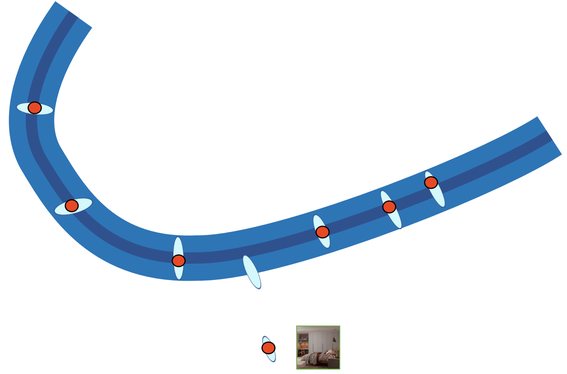
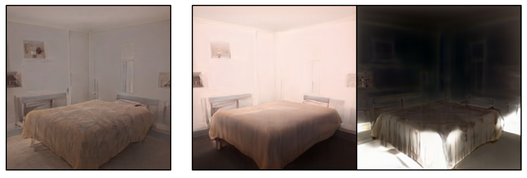
Target



Legend

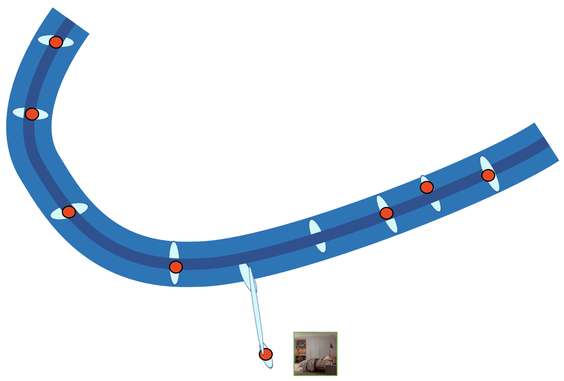


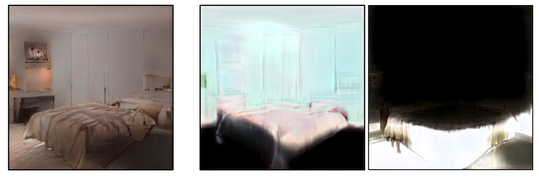


Inversion Edits

(a) Idinvert

- Inversion in latent style space ($\mathcal{W}^+$ space)
- Uses encoder with a fixed generator
- Struggles on inversion but preserves edit semantics

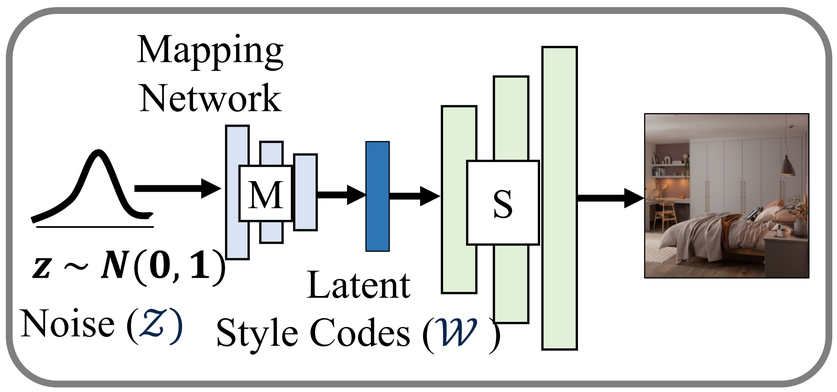




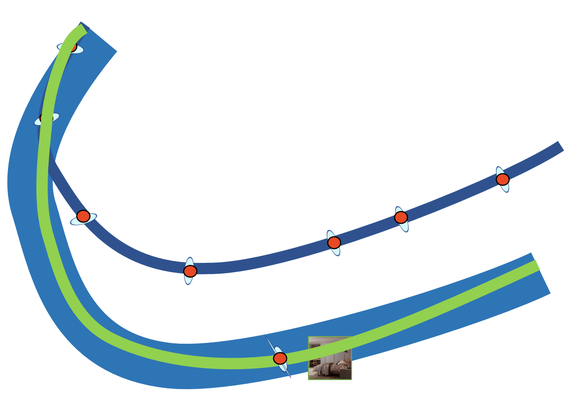
Inversion Edits

(b) StyleGAN Optimization

- Inversion in latent style space ($\mathcal{W}$ space)
- Performs direct optimization on fixed generator
- Struggles on inversion and loses editability



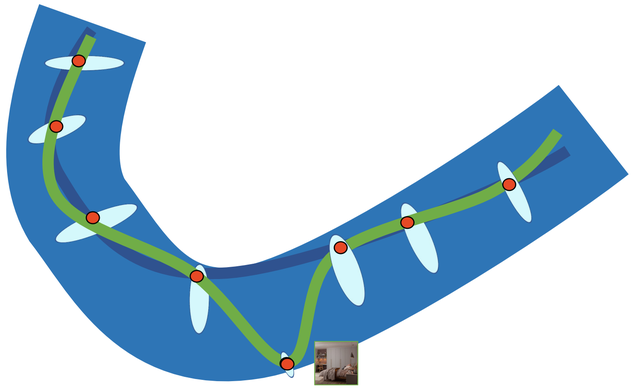
A StyleGAN Sketch Model



Inversion Edits

(c) Pivotal Tuning Inversion

- Inversion in latent style space (finds a pivot in $\mathcal{W}$ space)
- Fine-tune StyleGAN with the fixed pivot (w)
- Accurate inversion but imperfect edits
- Rotation of the edit space



Inversion Edits

(d) Make It So (Ours)

- Inversion in noise space ($\mathcal{Z}$ space)
- Performs joint optimization of z code and generator
- Steers the generator while preserving its characteristics
- Highly-accurate inversion with good editability